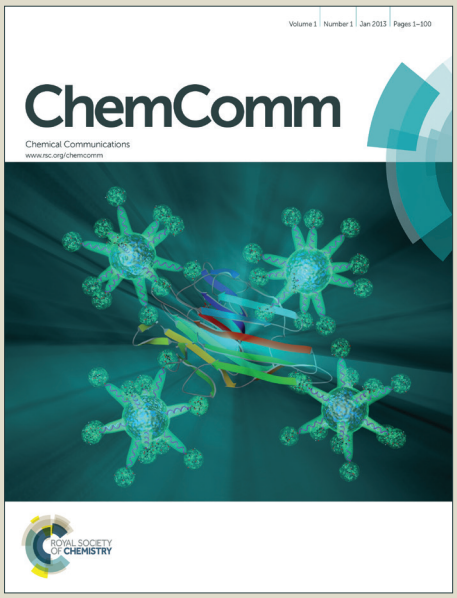
CrossMark
click for updates

View Article Online
View Journal

ChemComm

Accepted Manuscript

This article can be cited before page numbers have been issued, to do this please use: X. Huang, S. Wu, J. Zhou, J. Dai, S. Ma, C. Fu and C. Xue, *Chem. Commun.*, 2015, DOI: 10.1039/C5CC06111C.



This is an *Accepted Manuscript*, which has been through the Royal Society of Chemistry peer review process and has been accepted for publication.

Accepted Manuscripts are published online shortly after acceptance, before technical editing, formatting and proof reading. Using this free service, authors can make their results available to the community, in citable form, before we publish the edited article. We will replace this *Accepted Manuscript* with the edited and formatted *Advance Article* as soon as it is available.

You can find more information about *Accepted Manuscripts* in the Information for Authors.

Please note that technical editing may introduce minor changes to the text and/or graphics, which may alter content. The journal's standard Terms & Conditions and the Ethical guidelines still apply. In no event shall the Royal Society of Chemistry be held responsible for any errors or omissions in this *Accepted Manuscript* or any consequences arising from the use of any information it contains.

ROYAL SOCIETY OF CHEMISTRY

www.rsc.org/chemcomm

Please do not adjust margins

View Article Online
DOI: 10.1039/C5CC06111C

Journal Name

COMMUNICATION

TfOH-catalyzed domino cycloisomerization/hydrolytic defluorination of 2,3-allenyl perfluoroalkyl ketones

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

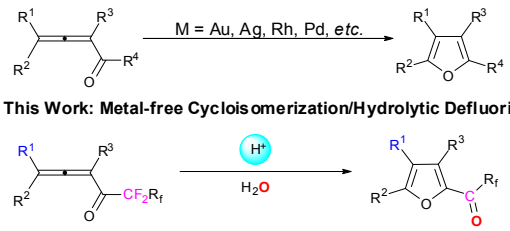
www.rsc.org/

Can Xue, Xin Huang, Shangze Wu, Jing Zhou, Jianxin Dai, Chunling Fu, and Shengming Ma*

A unique TfOH-catalyzed domino cycloisomerization/hydrolytic defluorination reaction of easily available *n*-perfluoroalkyl allenones in the presence of H_2O providing furanyl perfluoroalkyl ketones has been developed. The ^{18}O-labelling experiments confirmed that the oxygen atom of the carbonyl group in the final products originates from water.

Furans are ubiquitous heterocycles, widely existed in naturally occurring and biologically active compounds[1] and materials,[2] and have been applied as versatile building blocks in organic synthesis.[3] There are two general approaches commonly used for the preparation of these five-membered rings: functionalization of a pre-formed heterocyclic core[4] and assembly of the ring from acyclic precursors.[5] Considering the unique structure and substituent-loading capability, multi-substituted furans can be conveniently formed from transition metal-catalyzed cyclization (such as Au, Ag, Rh, Pd, *etc.*) of allenones[6-10] (Scheme 1): Marshall *et al.* reported the Ag^+- or Rh^+-catalyzed cycloisomerization of allenones to afford tri-substituted furans;[6] Hashmi and Alcaide found that Pd(II) was also a proper catalyst in this conversion;[7] Hashmi, Che, and later Alcaide developed Au-catalyzed cycloisomerization of allenones;[8] Gevorgyan's group had systematically studied the cascade 1,2-migration/cyclization of allenones substrates or intermediates under the catalytic of several transition metal catalysts.[9] Previously, we and Wang's group have demonstrated efficient synthetic routes for the preparation of multi-substituted furans via the Pd-catalyzed cross-coupling cyclization of 2,3-allenones with aryl halides, allylic halides or arylboronic acids, respectively.[10]

On the other hand, due to the mimic effect and very high electronegativity, the introduction of fluorine atoms into organic molecules would significantly change their properties, such as solubility, bioavailability, and metabolic stability.[11] Because of these special properties, fluorine containing compounds are of great importance and extensively applied in agricultural, pharmaceuticals, and materials science.[12] However, to the best of our knowledge there is no synthetic methodology reported for generating perfluoroalkyl substituted furans. Due to the fact that transition metal catalysts are usually expensive and toxic, here, we wish to report the metal-free TfOH-catalyzed[13] domino cycloisomerization of allenyl perfluoroalkyl ketones, which could be easily prepared according to the method developed in this group,[14] followed by hydrolytic defluorinaiton to yield furan-2-yl perfluoroalkyl ketones efficiently (Scheme 1).



Scheme 1. Cycloisomerization reactions of allenyl ketones.

We were studying the cyclization of easily available diaryl-substituted **1a** firstly with transition metal catalysts. Although perfluoroalkyl furan **2a** could be obtained in 46% NMR yield under the catalysis of $AuCl_3$ (entry 1, Table 1), no reaction occurred when AuCl, $Au(PPh_3)OTf$,[9g] $AgBF_4$,[9c] and AgOTf[9g,9h] were attempted as the catalysts (entries 2-5, Table 1). Furthermore, we explored such a conversion with different metallic trifluoromethanesulfonates, which provided us with very interesting results: when aged $Sc(OTf)_3$ was used as the catalyst, instead of perfluoroalkylfuran **2a**, an unexpected more polar product, which was finally identified as 4,5-diphenyl-3-propylfuran-2-yl perfluoropropyl ketone **3a** by the ^1H, ^{13}C, ^{19}F NMR, NOE, IR, MS and elemental analysis together with X-Ray single crystal diffraction data (Figure S1),[15] was obtained (entry 6, Table 1); neither newly purchased CuOTf·PhH nor

Laboratory of Molecular Recognition and Synthesis, Department of Chemistry
Zhejiang University
Hangzhou 310027, Zhejiang, People's Republic of China
E-mail: masm@sioc.ac.cn
Electronic Supplementary Information (ESI) available: Experimental details, ^1H, ^{13}C, ^{19}F and NOE NMR spectra. See DOI: 10.1039/x0xx00000x

Published on 22 September 2015. Downloaded by UNIVERSITY OF NEBRASKA on 22/09/2015 14:51:45.

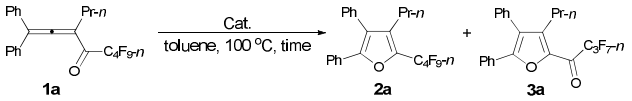
ChemComm Accepted Manuscript

View Article Online
DOI: 10.1039/C5CC06111C

Published on 22 September 2015. Downloaded by UNIVERSITY OF NEBRASKA on 22/09/2015 14:51:45.

$Cu(OTf)_2$ could catalyze this conversion successfully, with 97% and 98% recovery of **1a**, respectively (entries 7 and 8, Table 1); no reaction took place in the O_2 atmosphere (entry 9, Table 1); considering that the aged $Sc(OTf)_3$ may contain TfOH, we tried to add 1.5 equiv of H_2O to the reaction vessel. To our delight, 67 % of **2a** and 27% of **3a** were obtained (entry 10, Table 1). Then, TfOH was tested as the sole catalyst: using aged TfOH (10 mol%), which may also contain water, afforded a mixture of **2a** and **3a** (entry 11, Table 1).

Table 1. Unexpected formation of furanyl ketone **3a** from perfluoroalkyl allenyl ketone **1a**.[a]



1a 2a 3a

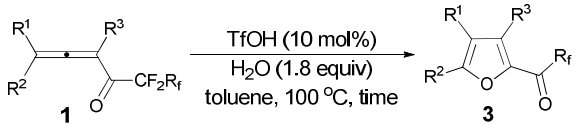
Entry	Cat. (mol%)	Time (h)	NMR Yield (%)[b]		Recovery of **1a** (%)[b]
			2a	**3a**	
1	$AuCl_3$ (10)	19	46	0	51
2	AuCl (5)	6	0	0	82
3[c]	$Au(PPh_3)OTf$ (2)	25	0	0	100
4[c,d]	$AgBF_4$ (5)	25	0	0	100
5[c,e]	AgOTf (20)	25	0	0	100
6	$Sc(OTf)_3$[f] (5)	11.5	0	57	0
7	CuOTf·PhH(10)	35	0	0	97
8	$Cu(OTf)_2$ (10)	12	0	0	98
9[g]	$Cu(OTf)_2$ (10)	10	0	0	87
10[h]	$Cu(OTf)_2$ (10)	10	67	27	0
11	TfOH[f] (10)	2	35	59	0

[a] The reactions were conducted on 0.2 mmol scale in toluene (2.0 mL). [b] Determined by the ^1H NMR analysis with CH_2Br_2 as the internal standard. [c] The reactions were conducted on 0.1 mmol scale in toluene (1.0 mL). [d] The reaction was conducted in DCE at 60 °C. [e] The reaction was conducted in DCM at rt. [f] Aged reagent. [g] The reaction was conducted within O_2 (1 atm provided by a balloon) atmosphere. [h] 1.5 equiv of H_2O were added.

Moreover, we varied the loading of water. Even in the absence of any added water, the reaction could still form **3a** in a decent yield (entry 1, Table S1), while a high loading of H_2O (up to 100 equiv) stopped the conversion (entry 9, Table S1). By comparing entry 2 with entry 3 of Table S1, reducing the dosage of TfOH to 5 mol% would yield a mixture of **2a** (88%) and **3a** (12%) at 60 °C with an extended reaction time as determined by the ^1H NMR analysis of the crude product. In the end, 1.8 equiv of H_2O came out to be the best choice (entry 5, Table S1).

With the optimized reaction conditions in hand, we then explored the scope of this transformation: the reaction of 1,1-diphenyl-substituted allenyl perfluoroalkyl ketones **1a-1c** proceeded to afford ketone **3** with excellent yields no matter R^3 is a propyl or a methyl group and R_f is a perfluoropropyl or a perfluoropentyl group (entries 1-3, Table 2). When a methyl group took the place of the phenyl group as R^2, the 1,2-Ph shift product **3d** was produced exclusively (entry 4, Table 2). Furthermore, 1,2-^{i}Pr migrated product **3f** was also formed (entry 6, Table 2). Interestingly, exclusive 1,2-^{i}Pr migration was observed affording **3e** in a moderate yield with R^1 = *iso*-propyl and R^2 = methyl (entry 5, Table 2). When R^1 = H and R^2 = Ph the tandem reactions could all proceed smoothly to afford 1,2-H shift products no matter R^3 was a propyl, an ethyl, or a methyl group (entries 7-10, Table 2). The substituent of the aryl group has almost no significant effect on the reaction (entries 11 and 12, Table 2). Albeit with a lower yield, 2-thienyl group may also be compatible in this conversion (entry 13, Table 2). To demonstrate the practicality of the procedure, a gram scale (3.0 mmol) reaction of **1g** was also conducted in moderate yield (63%) as shown in entry 8, Table 2.

Table 2. Domino cycloisomerization/hydrolytic defluorination reactions of **1**: the scope and selectivity of 1,2-shift.[a]

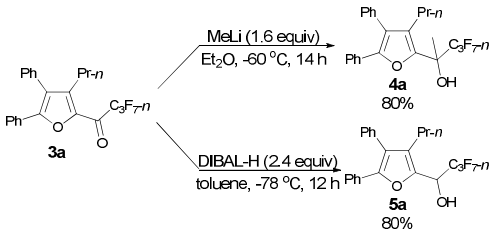


1 3

Entry	R^1	R^2	R^3	R_f	Time (h)	Yield of **3** (%)[b]
1[c]	Ph	Ph	*n*-Pr	*n*-C_3F_7 (**1a**)	2.5	96 (**3a**)
2[d]	Ph	Ph	*n*-Pr	*n*-C_5F_{11} (**1b**)	4	93 (**3b**)
3[d]	Ph	Ph	Me	*n*-C_3F_7 (**1c**)	6	94 (**3c**)
4	Ph	Me	Et	*n*-C_3F_7 (**1d**)	8	88 (**3d**)
5	*i*-Pr	Me	*n*-Bu	*n*-C_3F_7 (**1e**)	5	67 (**3e**)
6	*i*-Pr	*i*-Pr	*n*-Bu	*n*-C_3F_7 (**1f**)	10	50 (**3f**)
7	H	Ph	*n*-Pr	*n*-C_3F_7 (**1g**)	21	61 (**3g**)
8[e]	H	Ph	*n*-Pr	*n*-C_3F_7 (**1g**)	17	63 (**3g**)
9	H	Ph	Et	*n*-C_3F_7 (**1h**)	17	53 (**3h**)
10	H	Ph	Me	*n*-C_3F_7 (**1i**)	20	47 (**3i**)
11	H	*p*-MeC_6H_4	*n*-Pr	*n*-C_3F_7 (**1j**)	20	66 (**3j**)
12	H	*p*-FC_6H_4	*n*-Pr	*n*-C_3F_7 (**1k**)	17	60 (**3k**)
13[c]	H	2-Thienyl	*n*-Pr	*n*-C_3F_7 (**1l**)	17	39 (**3l**)

[a] Conditions: substrate (0.5 mmol), TfOH (10 mol%), H_2O (1.8 equiv), toluene (5 mL), at 100 °C. [b] Isolated yield. [c] The reaction was conducted on a 0.2 mmol scale in toluene (2.0 mL). [d] H_2O (2.2 equiv) was used. [e] The reaction was carried out on a gram scale (3.0 mmol).

The furanyl perfluoroalkyl ketone **3** producted here have been demonstrated to be very useful in organic synthesis as shown in Scheme 2: when **3a** was treated with MeLi in Et_2O at -60 °C for 14 h, perfluoropropyl containing tertiary alcohol **4a** could be obtained in 80% isolated yield[14] while secondary alcohol **5a** was afforded efficiently with DIBAL-H as the reducing reagent.



Scheme 2. Synthetic application of **3a**.

Considering that furanyl ketone **3a** may be formed in the presence of water and TfOH from **1a** (entry 6, Table 1), **2a** was

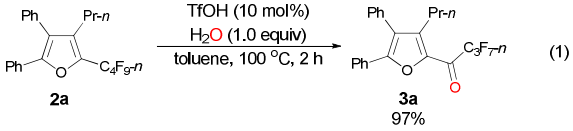
ChemComm Accepted Manuscript

This journal is © The Royal Society of Chemistry 20xx

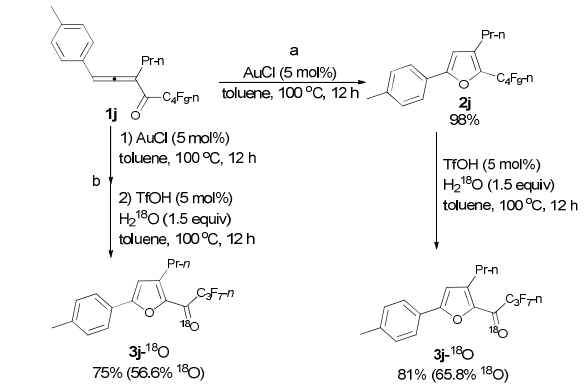
Please do not adjust margins
View Article Online
DOI: 10.1039/C5CC06111C

Published on 22 September 2015. Downloaded by UNIVERSITY OF NEBRASKA on 22/09/2015 14:51:45.

then chosen as the starting point. The reaction of **2a** with H_2O (1.0 equiv) and a catalytic amount of TfOH (10 mol%) in toluene at 100 °C for 2 h afforded **3a** successfully in 97% isolated yield (eq. 1). Thus, **2a** has been demonstrated to be the key intermediate to form **3a** from **1a**.



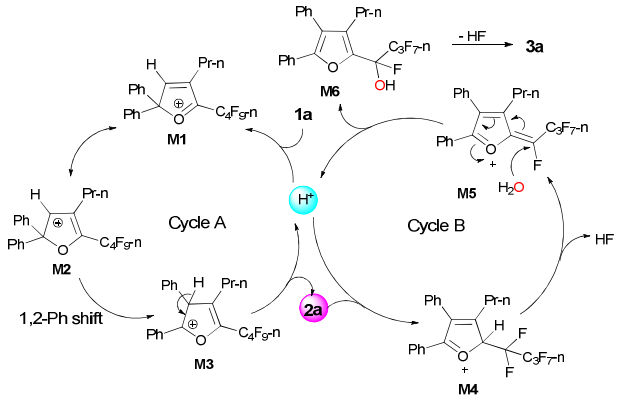
Furthermore, ^{18}O-labeling experiments were conducted: 2-perfluorobutyl-5-(4-methylphenyl)-3-propylfuran **2j** could be obtained in 98% isolated yield from **1j** with AuCl (5 mol%) in toluene at 100 °C for 12 h, then **2j** was treated with TfOH (5 mol%) and $H_2{}^{18}O$ (1.5 equiv) in toluene for 12 h to afford **3j**-^{18}O in 81% yield and 65.8% ^{18}O (Path a, Scheme 3); in the sequential two-step one-pot conversion, **1j** was treated firstly with AuCl (5 mol%) in toluene at 100 °C for 12 h, which was followed by the direct addition of TfOH (5 mol%) together with $H_2{}^{18}O$ (1.5 equiv) in toluene at 100 °C for another 12 h to provide **3j**-^{18}O in 75% yield with 56.6% ^{18}O for two steps (Path b, Scheme 3). Hence, it is confirmed that the oxygen atom of the carbonyl group originated from water.



Scheme 3. ^{18}O-labeling experiments.

Based on these results a possible mechanism is presented: In the presence of TfOH, the allene moiety of **1a** would interact with H^+ followed by nucleophilic attack from the carbonyl oxygen to form the cyclic oxonium intermediate **M1**. 1,2-Phenyl shift of the resonance structure **M2** would form **M3**. Aromatization would provide the tetra-substituted furan **2a** by releasing the catalytically active species H^+ (Cycle A, Scheme 4). The in-situ formed product **2a** would then react with H^+ in tandem to afford the oxonium intermediate **M4**, which would undergo the elimination of HF to afford **M5**. Subsequent nucleophilic attack of water with **M5** would form **M6**. Elimination of HF would finally yield **3a** (Cycle B, Scheme 4).[16]

In conclusion, we have identified H^+ for the first time as the catalyst for the novel domino cycloisomerization/hydrolytic defluorination of allenyl perfluoroalkyl ketones. In the presence of water, such a metal-free tandem TfOH-catalyzed reaction would efficiently conceive furan-2-yl perfluoroalkyl ketones, of which the carbonyl oxygen originated from water as confirmed by ^{18}O-labelling experiments. The selectivity of 1,2-shift has been established: H > Ar > secondary alkyl >primary alkyl. Considering the readily availability of the starting materials[14] and the potential of these perfluoroalkyl-containing furanyl ketones, this method will be useful in organic chemistry, materials sciences, and medicinal chemistry. Further studies in this area are still undergoing in our laboratory.



Scheme 4. Proposed mechanism with **1a** as the example.

Financial support from the National Natural Science Foundation of China (21232006) and the National Basic Research Program (2015CB856600) are greatly appreciated. Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University. The results presented in entry 7 of Table 3 have been kindly reproduced by Dr. Jiasheng Zhang.

Notes and references

1 For most important examples, see: (a) F. Bellina and R. Rossi, *Tetrahedron* 2006, **62**, 7213; (b) A. Fürstner and H. Weintritt, *J. Am. Chem. Soc.* 1998, **120**, 2817; (c) A. Fürstner, *Angew. Chem., Int. Ed.* 2003, **42**, 3582; (d) A. Fürstner, K. Reinecke, H. Prinz and H. Waldmann, *ChemBioChem* 2004, **5**, 1575; (e) A. Fürstner and E. J. Grabowski, *ChemBioChem* 2001, **2**, 706; (f) G. R. Pettit, J. McNulty, D. L. Herald, D. L. Doubek, J.-C. Chapuis, J. M. Schmidt, L. P. Tackett and M. R. Boyd, *J. Nat. Prod.* 1997, **60**, 180.

2 (a) Y. Miyata, T. Nishinaga and K. Komatsu, *J. Org. Chem.* 2005, **70**, 1147; (b) M. Heylen, K. van den Broeck, C. Boutton, M. van Beylen, A. Persoons and C. Samyn, *Eur. Polym. J.* 1998, **34**, 1453; (c) F. Ferrero, L. Napoli, C. Tonin and A. Varesano, *J. Appl. Polym. Sci.* 2006, **102**, 4121; (d) S. Venkatraman, R. Kumar, J. Sankar, T. K. Chandrashekar, K. Sendhil, C. Vijayan, A. Kelling and M. O. Senge, *Chem.-Eur. J.* 2004, **10**, 1423; (e) A. Facchetti, A. Abbotto, L. Beverina, M. E. van der Boom, P. Dutta, G. Evmenenko, G. A. Pagani and T. J. Marks, *Chem. Mater.* 2003, **15**, 1064; (f) L.-Z. Zhang, C.-W. Chen, C.-F. Lee, C.-C. Wu and T.-Y. Luh, *Chem. Commun.* 2002, 2336; (g) P. Novák, K. Müller, K. S. V. Santhanam and O. Haas, *Chem. Rev.* 1997, **97**, 207.

3 (a) H.-K. Lee, K.-F. Chan, C.-W. Hui, H.-K. Yim, X.-W. Wu and H. N. C. Wong, *Pure Appl. Chem.* 2005, **77**, 139; (b) H.

ChemComm Accepted Manuscript

View Article Online
DOI: 10.1039/C5CC06111C

Published on 22 September 2015. Downloaded by UNIVERSITY OF NEBRASKA on 22/09/2015 14:51:45.

N. C. Wong, P. Yu and C.-Y. Yick, *Pure Appl. Chem.* 1999, **71**, 1041; (c) B. H. Lipshutz, *Chem. Rev.* 1986, **86**, 795.

4 (a) D. S. Ennis and T. L. Gilchrist, *Tetrahedron* 1990, **46**, 2623; (b) T. R. Bailey, *Synthesis* 1991, 242; (c) S. Murai, F. Kakiuchi, S. Sekine, Y. Tanaka, A. Kamatani, M. Sonoda and N. Chatani, *Nature* 1993, **366**, 529; (d) P. Weyerstahl, A. Schenk and H. Marschall, *Liebigs Ann.* 1995, 1849. (e) J. Fournier, S. Arseniyadis and J. Cossy. *Angew. Chem. Int. Ed.* 2012, **51**, 7562; (f) C. A. Schoenbaum, D. K. Schwartz and J. W. Medlin, *Acc. Chem. Res.* 2014, **47**, 1438; (g) W. Wu and H. Jiang, *Acc. Chem. Res.* 2014, **47**, 2483.

5 For recent reviews, see: (a) N. T. Patil and Y. Yamamoto, *ARKIVOC* 2007, **x**, 121; (b) S. F. Kirsch, *Org. Biomol. Chem.* 2006, **4**, 2076; (c) R. C. D. Brown, *Angew. Chem., Int. Ed.* 2005, **44**, 850; (d) B. A. Keay, *Chem. Soc. Re*v. 1999, **28**, 209; (e) X. L. Hou, H. Y. Cheung, T. Y. Hon, P. L. Kwan, T. H. Lo, S. Y. Tong and H. N. C. Wong, *Tetrahedron* 1998, **54**, 1955; (f) S. Schröter, C. Stock and T. Bach, *Tetrahedron* 2005, **61**, 2245; (g) C. Schmuck and D. Rupprecht, *Synthesis* 2007, 3095. (h) Y. Xiao and J. Zhang, *Angew. Chem. Int. Ed.* 2008, **47**, 1903; (i) P. Lenden, D. A. Entwistle and M. C. Willis, *Angew. Chem. Int. Ed.* 2011, **50**, 10657; (j) C. He, S. Guo, J. Ke, J. Hao, H. Xu, H. Chen and A. Lei, *J. Am. Chem. Soc.* 2012, **134**, 5766; (k) B. Lu, J. Wu and N. Yoshikai, *J. Am. Chem. Soc.* 2014, **136**, 11598.
For transition metal-catalyzed cyclization of allenones, see ref. 6-10:

6 (a) J. A. Marshall and E. D. Robinson, *J. Org. Chem.* 1990, **55**, 3450; (b) J. A. Marshall and X.-J. Wang, *J. Org. Chem.* 1991, **56**, 960; (c) J. A. Marshall and G. S. Bartley, *J. Org. Chem.* 1994, **59**, 7169; (d) J. A. Marshall and C. A. Sehon, *J. Org. Chem.* 1995, **60**, 5966.

7 (a) A. S. K. Hashmi, *Angew. Chem. Int. Ed.* 1995, **34**, 1581; (b) A. S. K. Hashmi, T. L. Ruppert, T. Knöfel and J. W. Bats, *J. Org, Chem.* 1997, **62**, 7295; (c) B. Alcaide, P. Almendros and T. Martínez del Campo, *Eur. J. Org. Chem.* 2007, 2844.

8 (a) A. S. K. Hashmi, L. Schwarz, J.-H. Choi and T. M. Frost, *Angew. Chem. Int. Ed*. 2000, **39**, 2285; (b) C.-Y. Zhou, P. W. H. Chan, C.-M. Che, *Org. Lett.* **2006**, *8*, 325; (c) B. Alcaide, P. Almendros, J. M. Alonso and I. Fernández, *J. Org. Chem.* 2013, **78**, 6688.

9 (a) A. V. Kel'in and V. Gevorgyan, *J. Org. Chem.* 2002, **67**, 95; (b) J. T. Kim, A. V. Kel'in and V. Gevorgyan, *Angew. Chem., Int. Ed.* 2003, **42**, 98; (c) A. W. Sromek, A. V. Kel'in and V. Gevorgyan, *Angew. Chem., Int. Ed.* 2004, **43**, 2280; (d) A. W. Sromek, M. Rubina and V. Gevorgyan, *J. Am. Chem. Soc.* 2005, **127**, 10500; (e) T. Schwier, A. W. Sromek, D. M. L. Yap, D. Chernyak and V. Gevorgyan, *J. Am. Chem. Soc.* 2007, **129**, 9868; (f) A. S. Dudnik, V. Gevorgyan, *Angew. Chem., Int. Ed.* **2007**, *46*, 5195; (g) A. S. Dudnik, A. W. Sromek, M. Rubina, J. T. Kim, A. V. Kel'in and V. Gevorgyan, *J. Am. Chem. Soc.* 2008, **130**, 1440; (h) A. S. Dudnik, Y. Xia, Y. Li and V. Gevorgyan, *J. Am. Chem. Soc.* 2010, **132**, 7645.

10 (a) S. Ma and J. Zhang, *Chem. Commun.* 2000, 117; (b) S. Ma, J. Zhang and L. Lu, *Chem. Eur. J.* 2003, **9**, 2447; (c) S. Ma and L. Li, *Org. Lett.* 2000, **2**, 941; (d) Y. Xia, Y. Xia, R. Ge, Z. Liu, Q. Xiao, Y. Zhang and J. B. Wang, *Angew. Chem. Int. Ed.* 2014, **53**, 3917.

11 (a) J. T. Welch, *Tetrahedron* 1987, **43**, 3123; (b) J. Mann, *Chem. Soc. Rev.* 1987, **16**, 381; (c) J.-P. Bégué and D. Bonnet-Delpon, *Bioorganic and Medicinal Chemistry of Fluorine*, Wiley-Interscience, 2008; (d) D. O'Hagan, *Chem. Soc. Rev*. 2008, **37**, 308; and references therein.

12 (a) P. Jeschke, *ChemBioChem* 2004, **5**, 570; (b) S. Purser, P. R. Moore, S. Swallow and V. Gouverneur, *Chem. Soc. Rev.* 2008, **37**, 320; (c) W. K. Hagmann, *J. Med. Chem.* 2008, **51**, 4359; (d) P. Kirsch, *Modern Fluoroorganic Chemistry: Synthesis, Reactivity, Applications*, Wiley-VCH, Weinheim, 2004.

13 For reviews referred to Brønsted acid-catalyzed cyclization of allenes, see: (a) A. S. K. Hashmi, *Catal. Today* 2007, **122**, 211; (b) T. Cañeque, F. M. Truscott, R. Rodriguez, G. Maestri and M. Malacria, *Chem. Soc. Rev.* 2014, **43**, 2916.

14 G. He, C. Xue, C. Fu and S. Ma, *Synlett* 2010, 281.

15 Crystal data for compound **3a**: $C_{23}H_{17}F_7O_2$, *MW* = 458.37, Triclinic, space group P-1, Final *R* indices *[I > 2σ(I)]*, *R*1 = 0.1199, *wR*2 = 0.2977, *R* indices (all data) *R*1 = 0.1732, *wR*2 = 0.3230, *a* = 13.635(4) Å, *b* = 13.766(4) Å, *c* = 14.405(7) Å, α = 108.215(10)°, β = 105.015(10)°, γ = 110.595(7)°, *V* = 2187.5(14) Å^3, *T* = 296(2) K, Z = 4, reflections collected/unique: 22702/7576 (R_{int} = 0.0458), number of observations [>2σ(*I*)]3825, parameters: 577. CCDC: 993055.

16 For base catalyzed hydrolytic defluorination, see: (a) M. A. Ismail, H. H. Zoorob and L. Strekowski, *ARKIVOC* 2002, **x**, 1; (b) H. Lee, A. Czarny, M. A. Battiste and L. Strekowski, *J. Fluorine Chem.* 1998, **91**, 221; (c) X.-J. Yang, J.-T. Liu and F.-L. Zhao, *J. Fluorine Chem.* 2004, **125**, 415; (d) G. G. Furin, A. P. Krysin, N. I. Protsuk and V. A. Lopyrev, *Russ. J. Org. Chem.* 2006, **42**, 1429.

ChemComm Accepted Manuscript

This journal is © The Royal Society of Chemistry 20xx